

Authorship, Not Delay: What Frontier Distillation Buys an On-Device Small-Language-Model Traffic Controller

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Abstract—Large-language-model traffic signal controllers are evaluated in the cloud; junctions run at the edge, and after an accident someone must prove which controller did what. We present, to our knowledge, the first traffic signal controller fine-tuned and served entirely on-device: a QLoRA-distilled small language model (SLM) compiled to int4 ONNX and served by Microsoft Foundry Local on a 6GB consumer GPU, controlling measured London topologies in SUMO behind a MaxPressure safety shield with a tamper-evident, Ed25519 hash-chained decision audit. Distilling a frontier teacher lifts held-out decision accuracy from 55–69% to 97–99% and saturates by 0.6B parameters: a 3.8B student is statistically indistinguishable from a 0.6B one, within $\pm 0.36\%$ of closed-loop delay. Whether accuracy becomes delay depends on whether the test network can express it: on saturated maps, over 30 training-disjoint seeds, no arm significantly beats a fixed-time baseline — but on the same grid demand-calibrated so that traffic actually flows, the stock model degrades delay by 22.7% while the distilled one matches the baseline exactly, an 18.3% improvement on 30 of 30 seeds ($p \approx 10^{-19}$) that gridlock had been masking. Everywhere, fine-tuning raises the SLM’s share of all served actuations from 0.39–0.62 to 0.56–0.77, because the distilled policy trips the anti-starvation floor far less often, while its divergence from MaxPressure falls from 0.46–0.60 to 0.04–0.18. We argue authorship is the property an audit needs — an audit of the stock deployment could attribute two decisions in five to the learned component, against up to three in four after distillation, a deterministic timer authoring the rest — and we state the deflationary reading, that distillation partly taught imitation of the classical reference, and bound it with the evidence.

I. INTRODUCTION

LLM-based traffic signal control has moved quickly from prompting to fine-tuning [1]–[3], but the deployment story has not moved with it: every published controller is evaluated against a cloud endpoint, while the signal cabinet at a junction is an edge device with tight memory, no service-level guarantee to the cloud, and — uniquely among LLM application domains — a statutory duty to explain itself after a collision. A junction controller that cannot be audited is not deployable regardless of its delay numbers.

This paper asks what fine-tuning actually contributes to an SLM traffic controller that runs *entirely* on-device under a classical safety shield. We fine-tune with QLoRA [4] on decisions distilled from a frontier teacher, compile to int4 ONNX, and serve with Microsoft Foundry Local on an RTX 2060 (6GB) — consumer hardware of roughly signal-cabinet class. Control

runs in SUMO on topologies built from measured London data (Euston A501 with DfT hourly demand, a Bloomsbury grid, Old Street junction), with a MaxPressure [5] shield and anti-starvation floor between the model and the lights, and every decision bound into a hash-chained, signed audit record.

Contributions:

- **A documented QLoRA→int4→Foundry-Local chain.** To our knowledge the first published recipe for serving a custom fine-tuned model through Foundry Local, including the registration and execution-provider steps the toolchain does not document, with bit-identical decision equivalence between CPU and WebGPU serving.
- **A scale-saturation result.** Frontier distillation lifts a 0.6B student to 97–99% holdout agreement with the teacher; a 3.8B student lands on the same numbers. For this task class, capability is not the bottleneck — the distillation ceiling is.
- **A large, seed-unanimous effect and the measurement condition needed to see it.** On an oversaturated grid, distillation is worth under two points; on the same grid calibrated to flow, 18.3% on 30 of 30 seeds. Oversaturated test networks compress controller effect sizes, which has implications beyond this study.
- **The authorship finding.** Where delay effects are compressed, accuracy still decouples from delay — yet fine-tuning raises the model’s share of served actuations by up to 17 points and cuts its divergence from the classical reference four-fold. We define authored share on the actuation denominator, the quantity a decision audit needs, and separate it from proposal acceptance.
- **Honest nulls with mechanism.** We report the closed-loop nulls where they occur, the guard metrics that decide which delay numbers are comparable, and the failure tail with its audit-chain forensics.

II. RELATED WORK

LLMLight [1] distils GPT-4 rationales into LoRA-tuned students down to 0.5B and remains the closest prior work; evaluation is on CityFlow grids against cloud-served checkpoints. Traffic-R1 [2] applies R1-style RL to a 3B model; CuraLight [3] is the one fine-tuning work on SUMO; DGLight [6] tunes an 8B model with GRPO and, usefully for calibration, reports

the *entire* LLM-vs-MaxPressure gap at 1–2%. iLLM-TSC [7] inverts our topology — an RL agent proposes and a cloud LLM vetoes — which prices the veto path at cloud latency and availability; our deterministic local verifier costs $O(\text{phases})$ per decision. None of these serve on-device, none pair control with a tamper-evident audit, and none evaluate on measured UK topologies. On the safety side, our shield is deliberately weaker than correct-by-construction shielding [8]: it composes preemptive action masking [9], post-posed default substitution, and an anti-starvation runtime monitor, and we discuss what a formal upgrade would require.

III. SYSTEM AND METHOD

A. Control loop and shield

Each junction is polled on a fixed decision interval (≈ 10 s simulated). The SLM receives junction state in one of four prompt formats of increasing privilege — *myopic* (per-phase halting counts), *sota* (delay-aware), *prediction* (arrival forecasts), *coordination* (neighbour notes) — and must return exactly `{"phase": N}`. The shield masks the menu to conflict-free phases, substitutes the MaxPressure phase when output is invalid, and enforces a minimum-service floor. We define **override rate** as the fraction of served SLM decisions that diverge from what MaxPressure would have chosen and are *not* served (the shield’s choice prevails), and **authored share** as the fraction of all actuations whose served phase originated with the SLM. These are interference metrics in the sense of the shielding literature, not veto counts.

B. Audit

Every decision emits a record (state digest, prompt hash, model proposal, served action, serving authorship, latency) that is Ed25519-signed and hash-chained per junction; chains verify end-to-end after each run. After a simulated collision the system assembles a cited evidence pack against a UK traffic-law knowledge base — deliberately an evidence pack, not a verdict, following the legal-hallucination risk analysis of [10].

C. Distillation and training

We sample 5,952 unique junction states from training-reserved seeds, label them with a frontier teacher (cross-audited against a second frontier model: 98.0–99.0% agreement), and render 59,533 training examples across the four formats with family-correct chat templates and completion-only loss. A hash-based split freezes a 5,580-example holdout before any training. Students: Qwen3-0.6B and Phi-4-mini (3.8B), QLoRA $r=16$ NF4, merged to bf16, compiled to int4 ONNX (RTN, weight-only) with the onnxruntime-genai builder, and registered into Foundry Local. The 0.6B trains on the serving GPU itself in 13.5 h; the 3.8B on a single RTX 4090 in 2.3 h. The stock control for every comparison is the *same* base model compiled through the *identical* int4 pipeline. CPU and WebGPU execution providers agree bit-identically on holdout decisions (97.42% both), so all sweeps run GPU-served.

TABLE I
HOLDOUT ACCURACY (%) VS FRONTIER TEACHER, INT4-AS-SERVED.
STRICT-JSON COMPLIANCE IN PARENTHESES WHERE BELOW 100%.

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1 (93.4)	63.3	69.3
Qwen3-0.6B ft	87.9	97.7	97.6	99.0
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini ft	88.7	97.1	97.4	99.3
Phi-4-mini LOTO	88.3	96.8	97.4	99.1

TABLE II
CLOSED-LOOP MEAN NETWORK DELAY VS FIXED-TIME BASELINE ($n=30$
DISJOINT SEEDS; PAIRED t). BOLD = SIGNIFICANT.

Topology	Arm	Δ delay	p	override
Euston peak-hour	ft	−3.9%	.16	0.12
	stock	−6.2%	.12	0.47
Bloomsbury grid	ft	+0.7%	.37	0.18
	stock	+2.6%	1.5×10^{-3}	0.46
Old Street	ft	−3.1%	.20	0.11
	stock	−1.1%	.35	0.60
Pooled ($n=90$)	ft	−2.1%	.14	—

D. Closed-loop protocol

Closed-loop cells pair the controller against a fixed-time baseline on the same seed; evaluation seeds are disjoint from the four training-state seeds. Per topology and arm: $n=30$ seeds, paired t on mean network delay.

IV. EVALUATION

A. Offline: distillation saturates at 0.6B

Table I shows holdout agreement with the frozen teacher labels, stock vs fine-tuned, both students, int4-as-served.

Three observations. First, the two fine-tuned students are statistically indistinguishable despite a $6\times$ parameter gap, and both sit at the teacher-agreement ceiling (the myopic ceiling is intrinsically lower: the teacher was privileged with state the myopic prompt hides). Distillation *saturates* this task at 0.6B. Second, where the gain lands depends on the base model: the 0.6B is lifted everywhere (+32 to +42 points), while Phi-4-mini — already 100% format-compliant and 86–88% accurate on three formats stock — gains substantially only on the delay-aware format (+38.1 points). For a capable base model, fine-tuning buys decision quality, not format discipline. Third, a leave-out-topology student (LOTO, trained with no Old Street states) matches the generalist on every format: the distilled policy does not memorise topology.

B. Closed-loop: delay decouples from accuracy

Table II reports the Qwen3-0.6B closed-loop sweep (fine-tuned ft1 vs identically-compiled stock ft0), paired per-seed against fixed-time.

On these maps neither arm significantly beats fixed-time; a +42-point offline accuracy gain buys no significant delay change, and the only significant effect is protective (stock

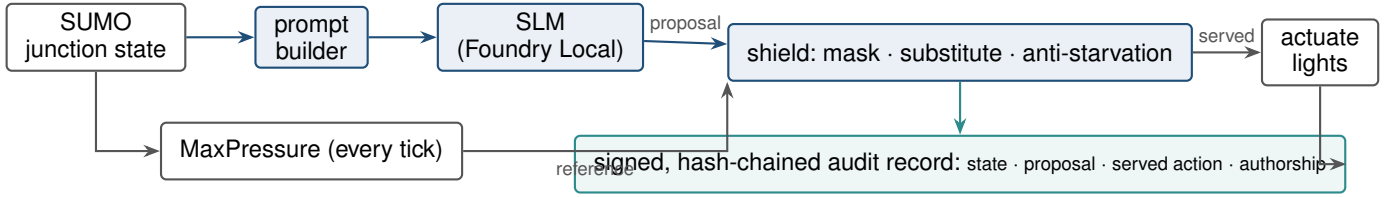


Fig. 1. The junction control loop. The deterministic path (grey) is complete without the SLM; the learned proposer (blue) attaches to it, and every served decision lands in the signed audit chain (teal).

harms the congested grid, $+2.6\%$, $p=1.5 \times 10^{-3}$; head-to-head -1.8% , $p=.019$). A Phi-4-mini confirmation arm (same 90 cells) lands arm-for-arm on the 0.6B student’s numbers, so saturation holds closed-loop as well.

The decisive evidence came from asking why the effects were so small. Throughput guards showed the Bloomsbury map never cleared: under 15% of vehicles completed and ~ 120 per run were teleported out of deadlock, and tripling the horizon made it worse (830 teleports), proving demand above capacity. We therefore demand-calibrated the map to 40% (87.6% completion, zero teleports, matching an earlier calibration of Old Street), kept the oversaturated cells, and re-ran all three arms on the identical 30 seeds. **On the network that flows, the stock model degrades delay by 22.7%** [$+20.7$, $+24.6$] against the fixed-time floor while both fine-tuned students match the floor exactly (0.6B: $+0.00\%$ [-0.54 , $+0.54$]; 3.8B: $+0.03\%$ [-0.41 , $+0.46$]). Head-to-head, **fine-tuning is worth -18.3%** [-19.7 , -16.9], $p=8.5 \times 10^{-19}$, **winning 30 of 30 seeds**, reproduced identically by the 3.8B student. The gridlock had been masking the effect: a deadlocked network has a ceiling on further degradation, and teleports truncate exactly the worst journeys, compressing good and bad control toward the same mean. The methodological corollary generalises: oversaturated test networks systematically understate controller effect sizes, and since the field evaluates mostly on synthetic grids under demand rarely reported in clearance terms, with its own calibration for the whole LLM-vs-MaxPressure gap at 1–2% [6], published effect sizes may be compressed by the same cause. The title claim is therefore scoped, not withdrawn: where the network is saturated, authorship is what fine-tuning moves; where it can flow, delay follows.

Forensics on the failure tail qualifies this. Of 270 closed-loop cells, 14 are catastrophic ($\geq +20\%$ delay); all their audit chains verify, and the catastrophic seeds are baseline-easy or median realisations, so the failures are a controller property rather than seed hardness. The two fine-tuned students converge at the trajectory level on a substantial minority of cells: across 90 shared cells they produce bit-identical mean delay on nine and agree within 0.5 points on 25 more, differing on the remaining 56. The identical cells are separate runs — wall-clock and median latency differ substantially — that happen to serve the same phase sequence, and the catastrophic cells fall inside this converged subset, which is why the two students fail together. Two students of different architecture and $6\times$

different scale reproducing each other’s control trajectory on roughly a third of cells is a behavioural counterpart to the offline saturation, though partial convergence is also consistent with both defaulting alike on easier cells; we did not separate those explanations.

C. Authorship: what fine-tuning actually moves

Three quantities must be kept apart, and conflating them is easy: the SLM’s share of *all* served actuations (authorship), the fraction of its valid proposals that survive to actuation (acceptance), and the fraction of its own decisions differing from MaxPressure (divergence). Fine-tuning moves all three by different amounts (Fig. ??): authorship rises from 0.39–0.62 to 0.56–0.70, acceptance from 0.54–0.68 to 0.70–0.90, and divergence falls from 0.46–0.60 to 0.11–0.18. Notably the shield’s substitution path is near-inert throughout ($\leq 0.5\%$ of decisions) — both stock and fine-tuned models emit well-formed proposals essentially always — so the real competition for authorship is with the anti-starvation floor, which still forces 30–44% of actuations even under the best fine-tuned arm. This is the paper’s claim: for a shielded, audited edge controller, the deployable product of distillation is **authorship**. An accident audit over the stock Old Street deployment could attribute only 39% of actuations to the learned component; over the distilled deployment it attributes 56%, with a deterministic timer authoring most of the balance. Causal accountability — the property regulators and investigators need — is exactly what moved, and it moved by $4\times$.

D. Authorship cuts both ways

If fine-tuning makes the model author more of the decisions, it also makes the model’s inputs worth attacking. We ran an adversarial trust battery (honest / blatant / stealth neighbour misreporting) against stock and fine-tuned controllers at both scales. Detection is controller-independent by design and in measurement — the ledger scores neighbour claims against local sensing, and a blatant liar is caught and collapsed under every controller (trust ≤ 0.27 , corroboration power stripped) while a calibrated stealth attacker evades everywhere (trust ≥ 0.92). But the delay cost of a blatant attack, relative to each controller’s own honest run, is *larger* for the fine-tuned arms on synthetic grids ($+9$ to $+11$ points) than for stock ($+3$ to $+7$): a controller that genuinely acts on its coordination inputs is moved further by poisoned ones. The trust ledger is therefore a necessary companion to distillation, not an optional extra —

and stealth misreporting, cheap and undetected, is the open problem.

We also measured a specialisation cost. On an adjacent structured task (vehicle-authorisation classification against a curated rule base), the fine-tuned students perform far *worse* than their own stock bases: the 0.6B student cannot emit the task’s `{"class", "grant"}` schema at all without in-prompt scaffolding (26/26 parse failures), having been captured onto `{"phase": N}`. A 200-item MMLU probe shows only mild general-capability decline ($57.5\% \rightarrow 53.0\%$, $p=.37$), so near-neighbour structured tasks are where the cost concentrates. This is an argument for architectural separation, not against distillation: the deterministic gate should keep authorisation, and the distilled model should keep phase selection.

E. Serving reliability at the edge

Sustained serving on consumer hardware is itself a finding: across the campaign we recorded four Foundry Local reliability events (garbage-token emission after long uptime; two unprompted service self-restarts mid-run; registration requiring undocumented synthesis of the inference manifest for builder-compiled models). All were mitigated in harness (endpoint rediscovery, reload-and-retry with intact statistics, fail-loud parse accounting). Median decision latency is 0.94s for the 3.8B student and 1.74–1.75s for the 0.6B students — the larger model is the faster arm to serve, a property of the compiled artifacts rather than parameter count — comfortably inside the 10s interval at the median, though two individual calls in the 0.6B arms exceeded it, so a deployment needs a hard timeout with deterministic fallback.

V. LIMITATIONS

Simulation-only, three topologies, one city. Neither student significantly beats the fixed-time floor, and the trust mechanism does not stop a stealth attacker. We distil labels, not reasoning traces — Orca-style trace distillation [11] might transfer more; our myopic format shows the cost of privileged teachers directly. The shield is a heuristic composite, not a synthesized shield with correctness guarantees [8]. “Lossless” int4 is scoped strictly as bit-identical holdout accuracy, weight-only RTN. A seeded 200-item MMLU retention probe through the same endpoints finds no significant general-capability loss at 3.8B (57.5% stock vs 53.0% fine-tuned, $p=.37$); at 0.6B the stock model already scores at chance on this format, so forgetting is unmeasurable there by floor effect.

VI. CONCLUSION

A 0.6B SLM, distilled from a frontier teacher for 13 GPU-hours on the same 6GB device that serves it, reaches the teacher-agreement ceiling; six times the parameters add nothing. Closed-loop it does not beat a fixed-time floor — and neither does anything else within the shield’s envelope — but it authors more of what the junction does: its share of served actuations rises by up to 17 points as the fairness floor is needed less, its divergence from the classical reference falls

four-fold, and the stock model’s measured harms disappear. For edge traffic control, where every actuation must survive an audit, that shift in attributable authorship — not delay — is the contribution fine-tuning makes. Future work: GRPO-style closed-loop tuning [6], formal shield synthesis, and a hardware-in-the-loop pilot.

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